AVIATOR GAME

UI/UX Analysis Document

Source: Spribe Official Aviator Demo

Analysis Date: March 30, 2026

1. CURVE / FLIGHT PATH ANIMATION

Shape & Mathematical Behavior

The curve follows an EXPONENTIAL growth pattern, not linear. The multiplier starts at 1.00x and accelerates rapidly as time progresses. The curve appears to follow a formula similar to: multiplier = e^(kt) where k is a growth constant.

OBSERVED BEHAVIOR: The curve starts relatively flat (gentle slope) at 1x-2x, then becomes progressively steeper. By 5x, the curve is climbing at approximately 45 degrees; by 10x+, it approaches near-vertical ascent.

Slope Progression at Key Milestones

- 1x-1.99x: Very gentle slope, approximately 10-15 degrees
- 2x-4.99x: Moderate slope, approximately 25-35 degrees
- 5x-9.99x: Steep slope, approximately 45-60 degrees
- 10x+: Very steep, approaching 70-80 degrees
- 50x+: Near-vertical, almost 85+ degrees

Derivative Behavior Near Crash

CRITICAL OBSERVATION: The curve does NOT flatten before crash. The plane maintains its exponential acceleration trajectory until the exact moment of crash. There is no plateau or deceleration warning - the crash happens instantaneously while the curve is still steeply ascending.

This creates the tension mechanic: players see the curve getting steeper and steeper, knowing it could crash at any moment while the potential reward increases exponentially.

Curve Visual Style

- Line Width: Approximately 3-4 pixels thick
- Color: Bright lime green (#86E96D) with glow effect
- Glow: Soft outer glow in same green color, approximately 8-10px radius
- Fill: Gradient fill under the curve from green to transparent
- Anti-aliasing: Smooth, high-quality curve rendering

Time to Reach Milestones (Typical Mid-Range Crash)

- 1x to 2x: Approximately 2-3 seconds
- 2x to 5x: Approximately 3-4 seconds
- 5x to 10x: Approximately 4-6 seconds
- Total to 10x: Approximately 10-12 seconds from start

Note: These timings vary based on the provably fair crash point generation. The video shows a 2.58x crash and an 11.18x crash.

2. PLANE SPRITE & ANIMATION

Plane Design Details

The plane is a stylized RED biplane/monoplane hybrid with the following characteristics:

- Fuselage: Sleek, streamlined red body with subtle gradient shading
- Wings: Single main wing (monoplane style) with white/light gray tips
- Tail: Classic vertical stabilizer with horizontal elevators
- Propeller: Visible at front, animated with motion blur effect
- Cockpit: Small, implied cockpit area
- Markings: No visible text or numbers on the plane itself

Color Specification

- Primary Red: Bright, saturated red (#E53935 or similar)
- Wing Tips: White or very light gray
- Propeller: Silver/gray with motion blur

- Shading: Subtle darker red gradient on underside

Bob/Floating Animation

The plane exhibits a subtle vertical bobbing motion independent of the curve movement:

- Amplitude: Approximately 5-10 pixels up/down
- Frequency: Approximately 2-3 complete cycles per second
- Pattern: Sine wave oscillation, smooth and natural
- Purpose: Adds life and dynamism to the sprite

Propeller Animation

- Animation: Continuous rotation
- Speed: Fast rotation, approximately 10-15 RPM visually
- Visual: Motion blur circle effect, not distinct blades
- Color: Silver/gray blur circle

Angle/Tilt Behavior

The plane rotates to follow the curve tangent:

- At 1x: Nearly horizontal, slight upward angle (~10 degrees)
- At 2x: Moderate climb angle (~25 degrees)
- At 5x: Steep angle (~45 degrees)
- At 10x+: Very steep angle (~60-70 degrees)

The rotation is smooth and follows the instantaneous slope of the curve, creating the illusion that the plane is 'flying' along the path.

Position on Screen

- Horizontal: Plane stays at approximately 60-70% from left edge
- Vertical: Plane moves upward as multiplier increases
- Camera: Follows the plane, panning up and right as curve ascends
- Trail: Particle trail follows behind the plane

3. FLY-AWAY / CRASH ANIMATION

Crash Behavior

At the moment of crash, the following sequence occurs:

- 1. Plane INSTANTLY disappears - no fade, no explosion
- 2. The curve stops drawing immediately
- 3. 'FLEW AWAY!' text appears above the final multiplier
- 4. Final multiplier turns RED and remains displayed
- 5. Background color shifts to darker purple/magenta

'FLEW AWAY!' Text Animation

- Position: Center of game area, above the multiplier
- Size: Large, approximately 48-64px font
- Color: White text (#FFFFFF)
- Font: Bold, sans-serif (likely system font)
- Animation: Fade-in from 0% to 100% opacity
- Duration: Text remains visible for approximately 3-5 seconds
- Style: All caps, with exclamation mark

Final Multiplier Display

- Position: Center, below 'FLEW AWAY!' text
- Color Change: Transitions from white to BRIGHT RED
- Size: Very large, dominant display (approximately 80-100px)
- Decimals: Always shows 2 decimal places (e.g., '2.58x')
- Duration: Remains visible until next round starts

Screen Effects

- Flash: NO bright flash at crash
- Shake: NO screen shake
- Color Overlay: Background shifts to darker purple/magenta tone

- Transition: Smooth fade to lobby state

Particle Effects

NO visible particle burst at crash point. The crash is a clean disappearance. However, during flight:

- Trail particles: Small white/yellow dots trailing behind plane
- Particle behavior: Fade out over time, create motion streak
- Density: Moderate, approximately 20-30 visible particles

4. MULTIPLIER DISPLAY

Position & Layout

- Position: Center of game canvas
- Anchor: Stays centered horizontally, moves up as curve climbs
- Z-Index: Above curve, below plane

Size & Scale

- Font Size: Approximately 15-20% of canvas height
- At 1x: Large, readable size
- At 10x+: Same size, no scaling with multiplier value

Color Behavior (State-Based)

- 1.00x - 1.99x: White (#FFFFFF)
- 2.00x - 9.99x: White (#FFFFFF) - no color change
- 10.00x+: Purple/Magenta tint on background, text remains white
- At Crash: Turns BRIGHT RED (#FF0000 or #E53935)

Font Specifications

- Typeface: Sans-serif, bold weight
- Style: Monospace-like numeric display
- Weight: Bold (700)
- Rendering: Anti-aliased, crisp edges

Animation Effects

- Number Update: Smooth increment, no jitter
- Decimal Animation: Hundredths place changes rapidly
- Pulse: Subtle pulse/glow effect at higher multipliers
- Scale: No scale animation, fixed size

Decimal Places

- Format: Always 2 decimal places
- Display: '1.00x', '2.58x', '11.18x'
- Suffix: Lowercase 'x' immediately after numbers
- Separator: Period (.) for decimal

5. BACKGROUND

Visual Style

The background features a dramatic SUNBURST/RADIAL RAY pattern emanating from the bottom-left corner:

- Pattern: Radiating rays like sunbeams or spotlight effect
- Origin: Bottom-left corner of canvas
- Direction: Rays extend toward top-right
- Count: Approximately 12-16 distinct rays

Color Scheme

- Base Color: Very dark blue/black (#0A0E17 or similar)
- Ray Color: Gradient from dark blue to lighter blue/purple
- Ray Brightness: Subtle, not overpowering
- Edge Softness: Rays have soft, blurred edges

Dynamic Background Changes

The background changes based on game state:

- Lobby/Waiting: Dark with subtle rays

- Flight (1x-2x): Blue-tinted rays
- Flight (2x-5x): Blue-purple gradient rays
- Flight (5x+): Purple/magenta tinted rays
- Crash: Dark purple/magenta dominant

Animation

- Ray Movement: Subtle rotation/pulse (very slow)
- Color Transition: Smooth gradient shifts during flight
- Intensity: Rays become more visible as multiplier increases

Grid Lines

- Presence: NO visible grid lines in the observed version
- Alternative: The curve itself provides the only visual reference

Note: Some Aviator variants may have subtle grid lines, but the official Spribe version observed uses the sunburst pattern exclusively.

6. BET PANEL UI

Layout & Position

- Position: Bottom of screen, fixed overlay
- Structure: Two identical bet panels side-by-side
- Width: Each panel approximately 45% of screen width
- Height: Approximately 25-30% of screen height
- Background: Dark semi-transparent overlay

Bet Amount Input

- Display: Large numeric field showing current bet amount
- Format: '1.00' with currency suffix
- Stepper Buttons: '-' and '+' on either side
- Quick Amount Buttons: 1, 2, 5, 10 (below input)

Tab Navigation

- Tabs: 'Bet' and 'Auto'
- Active Tab: Highlighted with underline
- Switching: Tap to toggle between manual and auto modes

BET Button (Green)

- Color: Bright lime green (#29A309)
- Text: 'Bet' + amount (e.g., 'Bet 1.00 USD')
- Shape: Rounded rectangle, full width of panel
- Height: Approximately 60-70px
- Font: White, bold, centered
- Hover: Slight brightness increase
- Press: Scale down slightly (0.95x)

CANCEL Button (Red)

- Color: Bright red (#D32F2F)
- Text: 'Cancel' + 'Waiting for next round'
- Shape: Same as BET button
- Purpose: Appears after bet is placed, allows cancellation

CASH OUT Button (Orange)

- Color: Orange/amber (#FF9800)
- Text: 'Cash Out' + current win amount
- Shape: Same as other buttons
- Dynamic: Amount updates in real-time with multiplier
- Purpose: Appears during flight to cash out active bet

Auto-Bet Controls

- Checkbox: 'Auto' toggle
- Auto Cash Out: Input field for target multiplier

- Visual: Slider or direct input

Balance Display

- Position: Top-right corner of screen
- Format: '30,000.88 USD'
- Color: Green (#29A309) for positive balance
- Updates: Real-time after wins/losses

7. SOCIAL / LEADERBOARD ELEMENTS

Left Sidebar Layout

A comprehensive betting information panel on the left side:

- Width: Approximately 25-30% of screen width
- Background: Dark semi-transparent
- Border: Subtle separator from main game area

Tab Navigation

- Tabs: 'All Bets', 'Previous', 'Top'
- All Bets: Shows current round's active bets
- Previous: Shows previous round results
- Top: Shows highest wins/leaderboard

Player Information Display

- Avatar: Colored circular avatar with initial/icon
- Username: Masked format (e.g., 'd***7', 'g***8')
- Bet Amount: Shows bet in USD
- Multiplier: Shows cash-out multiplier (if cashed out)
- Win Amount: Shows winnings in green if won

Player Count

- Display: '423/569 Bets' format

- Position: Top of sidebar
- Icon: Small person/group icon
- Updates: Real-time as players join/leave

Bet History Pills

- Position: Top of main game area, horizontal scroll
- Format: Small colored pills with multiplier values
- Color Coding: Green for low (1x-2x), Yellow/Orange for medium (2x-5x), Purple/Pink for high (5x+)
- Count: Shows last 15-20 rounds

Top Winners (Top Tab)

- Filter Options: 'Day', 'Month', 'Year'
- Display: Avatar, username, bet amount, result multiplier
- Win Amount: Large green numbers
- Round Max: Shows maximum multiplier reached

Chat Feature

- Presence: Chat panel visible on right side
- Toggle: Can be opened/closed
- Content: Player messages, system notifications
- 'Rain' Feature: Free bets dropped in chat

8. GAME STATE TRANSITIONS

State Machine Overview

The game follows a clear finite state machine: LOBBY -> BETTING -> FLYING -> CRASHED -> LOBBY

LOBBY State

- Visual: 'UFC Aviator OFFICIAL PARTNERS' logo displayed
- Background: Dark with subtle rays

- Multiplier: Not displayed
- Buttons: Green 'Bet' buttons active
- Duration: Variable, typically 5-10 seconds between rounds

LOBBY -> BETTING Transition

- Trigger: Timer countdown or manual bet placement
- Visual Change: 'Cancel' button appears after bet
- Text: 'Waiting for next round' appears
- Animation: Smooth fade transition

BETTING State

- Visual: Logo fades, plane prepares to appear
- Countdown: Brief countdown before flight
- Bet Status: 'Cancel' button shows bet is locked
- Duration: Approximately 3-5 seconds

BETTING -> FLYING Transition

- Trigger: Countdown reaches zero
- Plane Appearance: Plane fades in at bottom-left
- Curve Start: Curve begins drawing from (0,0)
- Multiplier: '1.00x' appears and starts increasing
- Button Change: 'Cancel' becomes 'Cash Out' (orange)
- Animation: Smooth 0.5 second transition

FLYING State

- Visual: Plane flying along curve
- Multiplier: Continuously updating
- Background: Color shifts with multiplier
- Cash Out: Button shows real-time win amount
- Duration: Variable, 2-30+ seconds depending on crash point

FLYING -> CRASHED Transition

- Trigger: Crash point reached (provably fair)
- Plane: Instantly disappears
- Curve: Stops drawing immediately
- Text: 'FLEW AWAY!' fades in
- Multiplier: Turns red, stops updating
- Duration: Instant crash, 3-5 seconds display

CRASHED -> LOBBY Transition

- Trigger: After crash display timeout
- Visual: Fade to UFC logo
- Reset: All bets cleared, new round setup
- Duration: Approximately 2-3 seconds
- Next Round: Setup begins immediately

9. SOUND CUES

Note: Sound analysis is limited as the video may not capture all audio clearly. The following is based on typical Aviator implementations and any audible cues:

Flight Start Sound

- Type: Engine start/whoosh sound
- Duration: Approximately 0.5-1 second
- Character: Rising pitch, energetic

Multiplier Tick Sounds

- Type: Subtle tick/click at each decimal update
- Frequency: Increases with multiplier speed
- Character: Light, rhythmic

Cash Out Sound

- Type: Success chime/cash register sound
- Character: Positive, rewarding tone
- Duration: Short, approximately 0.3 seconds

Crash Sound

- Type: Engine cut-off/descent sound
- Character: Sudden stop, slightly negative
- Duration: Very short, approximately 0.2 seconds

Settings

- Sound Toggle: Available in settings menu
- Music Toggle: Separate background music control
- Animation Toggle: Can disable plane animation

10. MICRO-INTERACTIONS

Button Hover States

- Green Bet Button: Brightness increases ~10%
- Orange Cash Out: Glow effect intensifies
- Red Cancel: Subtle highlight
- Timing: Immediate response (<50ms)

Button Press Animations

- Scale: Button scales to 0.95x on press
- Color: Slight darkening during press
- Release: Springs back to 1.0x
- Duration: Approximately 100ms

Bet Confirmation

- Visual: Button changes from green to red

- Text: Changes from 'Bet' to 'Cancel'
- Sub-text: 'Waiting for next round' appears
- Animation: Smooth color transition (200ms)

Cash Out Animation

- Button: Orange button press animation
- Win Display: Green 'You have cashed out!' banner
- Amount: Shows win amount with multiplier
- Balance: Updates with animation
- Banner Duration: Approximately 3 seconds

Win Amount Display

- Position: Top banner overlay
- Color: Green background, white text
- Content: 'You have cashed out!' + multiplier + win amount
- Dismiss: Manual X button or auto-dismiss

Settings Menu Interactions

- Open: Slide-in from right
- Close: Slide-out to right
- Toggles: Smooth switch animation
- Duration: 300ms transitions

History Pill Interactions

- Hover: Slight scale increase
- Click: Opens detailed round info
- Scroll: Horizontal smooth scroll

Premium Polish Details

What makes the Spribe Aviator feel premium:

- Smooth 60fps animations throughout

- Consistent easing curves (ease-out for transitions)
- Subtle glow effects on active elements
- Color-coded feedback (green=win, red=loss, orange=action)
- Real-time updates with no visible lag
- Responsive button states
- Clean, uncluttered UI layout
- Professional typography hierarchy
- Seamless state transitions
- Visual excitement that builds with multiplier